



Motilal Nehru National Institute of Technology Allahabad, Prayagraj

Department of Computer Science and Engineering

Introduction to Artificial Intelligence and Machine Learning Lab

(CSN12601)

Lab Assignment – 6

S. N.	Questions
1.	Apply the following data pre-processing and baseline modeling steps on the Titanic dataset: a) Write Python code to load the Titanic CSV file into a DataFrame. b) Write code to detect and report missing values in each column. c) Write code to drop irrelevant/unnecessary columns from the dataset. d) Write code to encode the Sex column, converting 'male' → 1 and 'female' → 0. e) Write code to handle missing values in the Age column (e.g., impute using mean/median). f) Write code to separate input features (X) and target output (y). g) Write code to split the dataset into training and testing sets (train-test split). h) Initialize and train a Logistic Regression model on the training data. i) Write code to test the Logistic Regression model and compute performance metrics (e.g., accuracy, confusion matrix, precision, recall, F1-score). j) Initialize and train a Gaussian Naive Bayes model on the training data. k) Write code to test the Naive Bayes model and compute performance metrics (e.g., accuracy, confusion matrix, precision, recall, F1-score). Dataset Link: https://github.com/datasciencedojo/datasets/blob/master/titanic.csv